Governing Simulator Discrepancy via Deterministic Pairing and Horizon-Validated Residual Dynamics

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Abstract

Residual learning across simulators is ill-posed if trajectory pairing, state semantics, and rollout timing are not aligned. In robotics pipelines, this creates a silent failure mode: models can look correct on one-step logs but fail dynamically under contact-rich execution. We present a governance-oriented sim-to-sim pipeline for PyBullet–MuJoCo Franka pushing that enforces deterministic pairing at episode/timestep granularity, applies a hard alignment gate before training, and evaluates residual correction under long-horizon stress. The correction is formulated as $\hat{S}_{t+1} = \Pi(S^f_{t+1} + \Delta(S_t, A_t))$, where Π maintains valid angles, quaternions, and bounded velocities. We report physical-state and contact-proxy metrics separately to avoid conflating solver-dependent bookkeeping with physically meaningful mismatch. On the latest passing hard-mode run, physical one-step p95 decreases from 0.1196 to 0.0207, and corrected rollout p95 remains stable at horizons 50/200/500 (0.0173/0.0174/0.0222) with zero NaN/Inf/divergence. Contact-regime physical p95 remains low (0.0075 contact, 0.0131 near-contact, 0.0082 force-top-5%). These results support deterministic pairing + alignment gating as practical controls for stable sim-to-sim residual correction under contact.

1 Introduction

Robotics teams routinely combine simulators to balance throughput, contact fidelity, and engineering convenience. The practical risk is not only model bias, but governance failure: simulator disagreement can enter training silently when reset conventions, frame semantics, or temporal indexing are not verified. As a result, residual models may appear accurate per-step while failing when executed over horizon.

This work treats cross-simulator correction as a systems validation problem. Rather than assuming paired data is valid, we enforce deterministic plan replay and explicit sample correspondence, then block training with a hard gate when alignment checks fail. We further require horizon-level validation in contact-rich regimes, because one-step loss alone is not a sufficient criterion for dynamic correctness.

Contact interactions are the stress case where hidden mismatch is most likely to surface. Near-contact and high-force phases amplify small semantic differences in collision handling, velocity integration, and state extraction. These effects may be negligible in free-space but accumulate under rollout, making contact slices and long-horizon checks central to trustworthy evaluation.

Figure 1 summarizes the governance pipeline used in this paper.

Our contributions are:

1. A deterministic paired-rollout protocol with strict episode/time index matching.
2. A hard alignment gate that blocks training when reset, timing, or frame consistency fails.

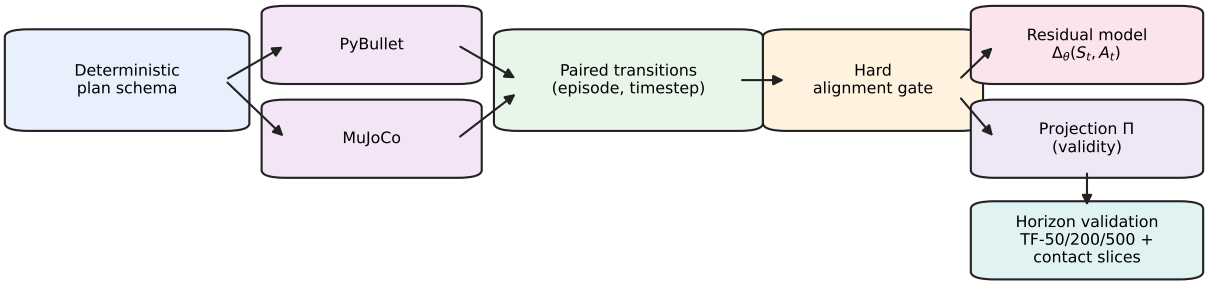

Figure 1: Governance pipeline for sim-to-sim residual correction: deterministic plan replay, paired transitions, hard alignment gate, projected correction, and horizon/contact validation.

3. A residual correction model evaluated under one-step, 50/200/500-step rollouts, and contact-focused slices.

4. A reporting protocol that separates physical-state error from solver-dependent contact-proxy error.

2 Related Work

Prior sim-to-real and sim-to-sim work uses domain randomization, system identification, and residual learning [5, 4, 2, 3]. Recent theoretical work on geometry, flatness, and information concentration under structured designs also informs how we think about stability and constrained correction [1]. These approaches are powerful but often under-specify cross-simulator *validation semantics*: paired-index integrity, one-step vs rollout mismatch, and contact-regime stability are frequently conflated.

Our focus is orthogonal: we provide a deterministic measurement-and-gating protocol around residual correction, then evaluate under explicit long-horizon and regime-sliced stress conditions.

3 Problem Formulation

Let $S_t \in \mathbb{R}^n$ and $A_t \in \mathbb{R}^m$ denote state and action. We use two simulators: fast simulator $\mathcal{E}_f$ and reference simulator $\mathcal{E}_r$. For paired transitions,

$$(S_t, A_t, S^f_{t+1}, S^r_{t+1}). \tag{1}$$

We learn a residual model

$$\Delta_\theta(S_t, A_t) \approx S^r_{t+1} - S^f_{t+1}, \tag{2}$$

and corrected next state

$$\hat{S}_{t+1} = \Pi\left(S^f_{t+1} + \Delta_\theta(S_t, A_t)\right), \tag{3}$$

where Π is a projection operator that enforces state validity.

Projection Π. We apply quaternion renormalization, joint-angle wrapping to $[-\pi, \pi]$, bounded velocity clipping, and bounded contact metadata clipping. This is used both in training and rollout evaluation.

4 Deterministic Pairing and Alignment Gate

4.1 Deterministic paired data

Both simulators execute the same deterministic plan schema with shared episode IDs and per-step timestamps. Pairing validity requires: identical sample count, episode monotonicity, and unit timestep increments.

4.2 Hard alignment gate

Training is blocked unless alignment checks pass:

- reset consistency under shared plan,
- effective step-time consistency,
- frame-consistent state extraction,
- strict early-step drift tolerance.

5 Residual Models

We evaluate a conditioned mixture-of-experts residual model with regime-aware gating. Inputs are (S_t^f, A_t) plus family and parameter conditioning metadata. Training uses robust Huber loss on corrected next state, guarded gradients, and projection-consistent updates.

6 Evaluation Protocol

6.1 Metric spaces

We report in original units with disjoint masks:

- **Physical-state dims**: robot, end-effector, object pose/velocity.
- **Contact-proxy dims**: contact flag, contact force proxy, min distance.

Pass/fail uses physical-state metrics only; contact-proxy metrics are separate diagnostics. If contact proxies are non-finite or contact-flag agreement collapses, verdict fails.

6.2 Stress protocol

- One-step physical error (mean/p95/max).
- Teacher-forced rollout physical error at horizons 50/200/500.
- Free-running stability rollout (NaN/Inf/divergence checks).
- Regime slices: contact, near-contact, force-top-5%, action quantiles, temporal slices.

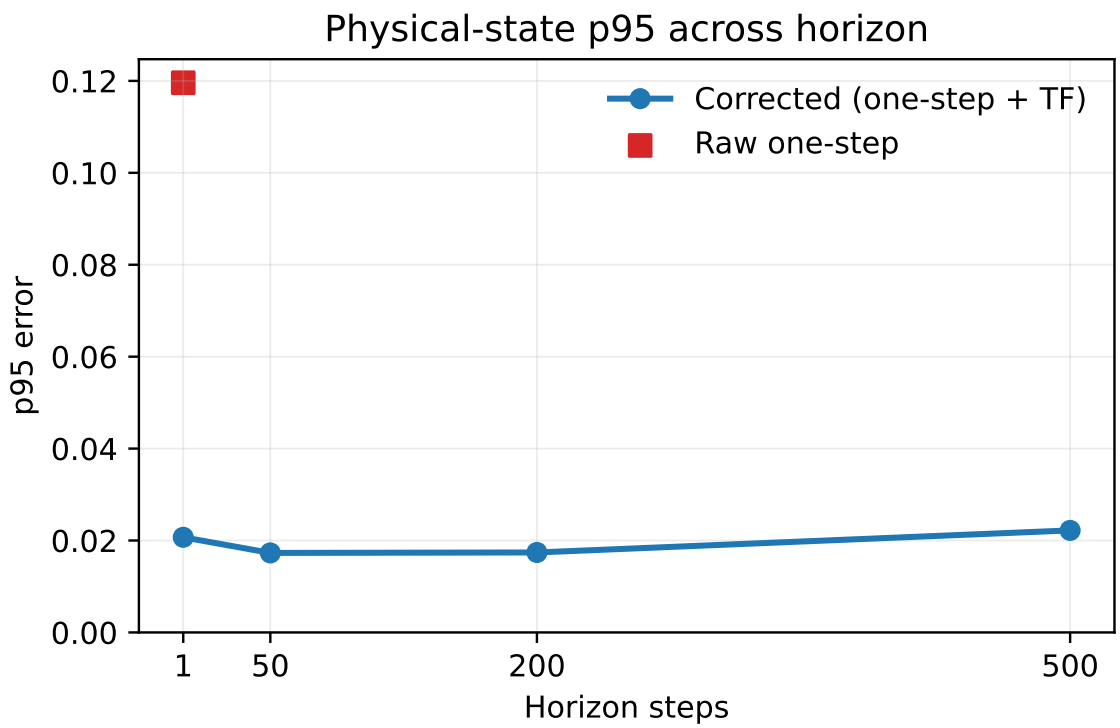

Figure 2: Physical-state p95 vs horizon. Corrected error remains low from one-step to 500-step teacher-forced rollout.

- Distribution shift: seed holdout (D1) and family holdout (D2).

Figure 2 visualizes horizon-wise p95 behavior, and Figure 4 summarizes contact-slice p95 values.

7 Results

Table 1 reports the final hard-mode run (PASS) from `reports/stress_20260203-173231`.

Figure 3 complements Table 1 with the publication-style summary plot used in the curated release bundle. It makes the one-step-to-rollout stability story visible at a glance instead of requiring the reader to infer it only from the table.

Contact-regime physical slices.

- contact1 p95: 0.0075
- near-contact p95: 0.0131
- force-top-5% p95: 0.0082

All required hard-mode thresholds pass.

Contact-proxy diagnostics (separate). Contact-flag accuracy is 0.9911; proxy p95 values are bounded (force: 0.1236, min-dist: 0.0046).

Table 1: Main physical-state results (original units).

Metric	Mean	p95	Max
Raw one-step (PHYS)	0.0941	0.1196	0.4627
Corrected one-step (PHYS)	0.0106	0.0207	0.1006
Rollout TF-50 (PHYS)	0.0100	0.0173	0.0844
Rollout TF-200 (PHYS)	0.0099	0.0174	0.0821
Rollout TF-500 (PHYS)	0.0110	0.0222	0.0872

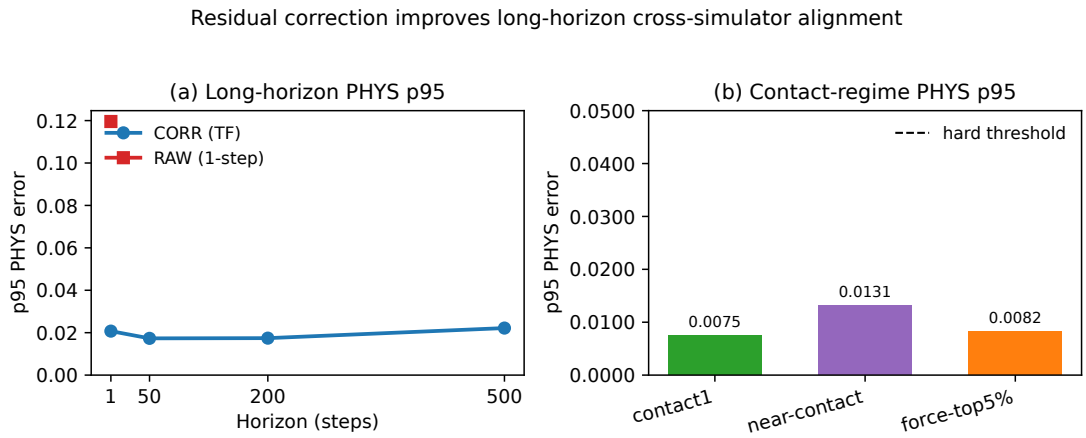

Figure 3: Canonical rollout summary figure from the passing hard-mode run. Raw one-step physical error is reduced sharply, and corrected teacher-forced rollout error remains low through horizons 50, 200, and 500.

Distribution shift. Both D1 seed holdout and D2 family holdout pass in hard mode. Across all reported D2 families in the stress report, corrected physical p95 remains tightly clustered between 0.0198 and 0.0214, which supports the claim that the learned correction is not carried by a single easy plan family.

Stability. Free-running stability rollouts at 50/200/500 steps show zero NaN/Inf/divergence.

Behavioral acceptance evidence. The public proof package also includes a separate behavioral PASS run designed for visual inspection rather than hard-mode stress. All behavioral gates pass: determinism, state set/get identity, settle stability, motion, cross-simulator agreement, one-step improvement, correction visibility, rollout improvement, acrobatics, and artifact generation. The selected run reaches an end-effector distance to target of 8.39×10^{-5}, Bullet-vs-MuJoCo terminal mismatch of 0.0329, and a best visual correction mixing coefficient of 0.005.

8 Discussion

The dominant practical insight is semantic consistency: projection-corrected residual models can remain stable only when state semantics, pairing, and evaluation loops are explicitly aligned. Without this governance layer, one-step gains can coexist with rollout failure. The broader report data strengthens that point. The strongest action and temporal stress slices still improve by factors

Table 2: Generalization and stability checks from the hard-mode PASS run.

Check	Count / scope	Mean	p95
D1 seed holdout	176160 samples	0.0106	0.0207
D2 family holdout: edge_push	114480 samples	0.0110	0.0198
D2 family holdout: near_contact_skims	169200 samples	0.0097	0.0214
D2 family holdout: stop_go	110880 samples	0.0081	0.0201
TF-50 rollout (PHYS)	200 starts	0.0100	0.0173
TF-200 rollout (PHYS)	200 starts	0.0099	0.0174
TF-500 rollout (PHYS)	200 starts	0.0110	0.0222
Free-running stability H50/H200/H500	200 starts each	0 NaN/Inf	0 divergence

Table 3: Representative worst-slice and regime-slice physical p95 values.

Slice	Raw p95	Corrected p95	Improve ratio
contact1	0.1004	0.0075	20.79
near-contact	0.1151	0.0131	13.44
force-top-5%	0.4626	0.0082	26.71
t_0_20	0.2053	0.0358	7.06
A_q00_25	0.1949	0.0289	9.66
far-contact	0.1501	0.0236	7.28

between roughly 7× and 27×, and the holdout families remain tightly bounded despite substantial variation in contact regime and motion style. This matters more than a single average number because the failure mode in sim-to-sim correction is usually regime-local rather than uniformly distributed.

The visual PASS run should not be conflated with the hard-mode quantitative run, but it does answer a different and important question: whether the public artifact shows coherent, deterministic, inspectable behavior rather than a metric-only claim. Taken together, the stress suite and the behavioral suite support a narrower but more defensible conclusion than a single benchmark table alone.

9 Limitations

This work is sim-to-sim only, on one robot/task family (Franka push), with deterministic plan generation. We do not claim sim-to-real transfer.

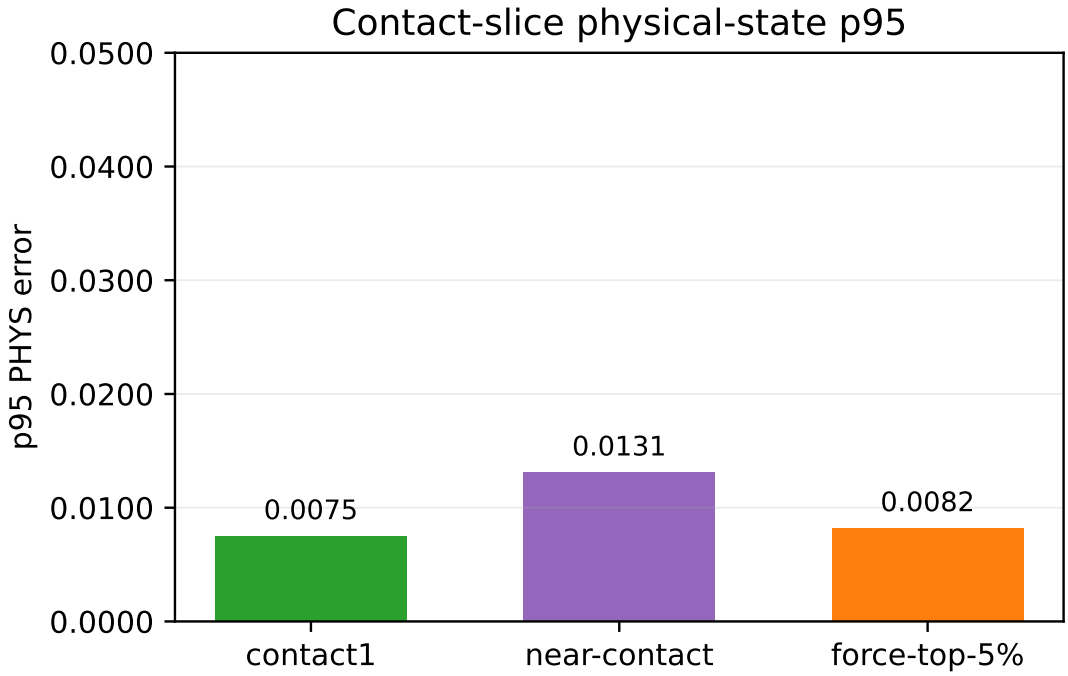

Figure 4: Contact-slice physical-state p95 values remain low across contact1, near-contact, and force-top-5% regimes.

10 Conclusion

Simulator mismatch can be governed through deterministic pairing, alignment gating, projection-consistent residual correction, and horizon-validated stress testing. Under this protocol, cross-simulator correction remains accurate and stable in contact-rich long-horizon settings.

References

[1] O. Anthony. A universal curvature–information principle: Flatness and d^{-1} concentration under 2-designs, 2025.

[2] Stephen James and et al. Sim-to-real via sim-to-sim: Data-efficient robotic grasping via randomized-to-canonical adaptation networks. *CVPR*, 2019.

[3] Fabio Muratore and et al. Data-efficient domain randomization with bayesian optimization. *IEEE Robotics and Automation Letters*, 2021.

[4] Xue Bin Peng, Marcin Andrychowicz, Wojciech Zaremba, and Pieter Abbeel. Sim-to-real transfer of robotic control with dynamics randomization. *ICRA*, 2018.

[5] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *IROS*, 2017.

Table 4: Behavioral acceptance metrics from the curated visual PASS run.

Behavioral check	Result	Key metric
Determinism	PASS	identical replay hashes
Set/get identity	PASS	state restoration preserved
Settle stability	PASS	raw jump 0.00339, corrected jump 0.00338
Motion / task completion	PASS	ee distance to target 8.39×10^{-5}
Cross-sim agreement	PASS	terminal Bullet-vs-MuJoCo distance 0.0329
Correction visibility	PASS	corr panel variance 2742.21
One-step improvement	PASS	corrected mean 0.03276 vs raw 0.03284
Correction rollout improves	PASS	best alpha 0.005
Acrobatics / artifacts	PASS	triptych and replay bundle generated

A Reproducibility Artifacts

Primary passing run: `reports/stress_20260203-173231`.
Model: `reports/delta_train_hardgen_fix_20260203-135301/model.pt`.
Datasets: `data_pybullet/paired_bullet_hardgen_seeds0-5.npz`, `data_mujoco/paired_mujoco_hardgen_seeds`

Additional packaged evidence. Behavioral acceptance anchor: `reports/20260210-173200/behavioral_acce`
Canonical rollout summary figure: `figures/rollout_phys_p95.pdf`.
Pipeline and contact figures: `figs/pipeline.pdf`, `figs/p95_horizon.pdf`, `figs/contact_slices_p95.pdf`.
Curated public proof bundle: triptych video, standalone Bullet replay, standalone MuJoCo replay, and overlay plots described in the public release documentation.